MATRIX Remote QM Execution on oracle

Running Gaussian, Molpro and ORCA jobs for MATRIX workflows

MATRIX Development Notes

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Abstract

This manual describes the operational contract for running external quantum chemistry programs on the private Linux host `oracle` and bringing the completed results back into `MATRIX`. The remote machine is used only for execution of heavy QM jobs. `MATRIX` remains the owner of the molecular state, shared `xyzin` sections, generated Gaussian GIC inputs and normalized post-processing through the single QM-adapter layer.

Contents

1 Scope

The `oracle` machine is intended for long background calculations with Gaussian, Molpro and ORCA. It is not a replacement for the local `MATRIX` source tree and it must not become a second set of parsers. The normal workflow is:

1. prepare or validate the molecule locally with `MATRIX`;
2. generate the QM input locally when `MATRIX` owns the input format, for example Gaussian ReadAllGIC files from frozen NEO/GICForge data;
3. copy the input file to `oracle`;
4. submit the job in the remote `~/matrix` work area;
5. copy completed logs, checkpoint-derived files or Molden files back;
6. promote the output through the corresponding `MATRIX` adapter into shared `xyzin` sections.

Passwords, personal credentials and site-specific secrets must not be written in the repository, in job inputs or in shell scripts. Use the interactive SSH prompt or a normal SSH key managed outside the repository.

2 Remote Directory Layout

The remote execution area is `~/matrix`. The following subdirectories have fixed meanings:

Directory	Purpose
`~/matrix/bin`	MATRIX helper scripts on the remote host.
`~/matrix/inputs`	Input files copied from the local workstation.
`~/matrix/jobs`	Per-job working directories created by the submit helper.
`~/matrix/logs`	Lightweight submit logs and redirected program output.
`~/matrix/outputs`	Optional collection area for selected final products.
`~/matrix/repo`	Optional clone or mirror of MATRIX when needed for tests.
`~/matrix/scratch`	MATRIX-owned scratch area; large Gaussian scratch is controlled separately.

Gaussian scratch is configured through `GAUSS_SCRDIR`. The current policy is to keep large Gaussian scratch under the local scratch filesystem configured on `oracle`, not inside the Git repository.

3 Environment

The login shell on `oracle` sources `~/.matrix_env.sh` from both `~/.bashrc` and, when available, `~/.zshrc`. The environment file is intentionally narrow: it exposes only the production Gaussian choices and the supported module stack.

```
setgau gdv32
setgau g16
module load molpro/2025.3
module load orca/6.1.1
```

Older GDV variants are disabled for MATRIX use. Jobs must use only `gdv32` or `g16`. Do not source historical external-code environments such as `elecext-env`, `External` or `RunGauExternal`; they are outside the MATRIX execution contract.

Useful checks after login are:

```
command -v gdv
command -v g16
command -v molpro
command -v orca
module list
echo "$GAUSS_SCRDIR"
```

The expected Molpro version for this machine is Molpro 2025.3. If `module load molpro/2025.3` stops working, treat that as an environment configuration problem and do not silently fall back to an older Molpro module.

4 Direct Program Commands

The direct commands are useful for small checks and for debugging the environment.

4.1 Gaussian GDV32

```
setgau gdv32
gdv32run molecule.gjf
```

Use GDV32 for Gaussian jobs that need the current group-specific Gaussian features. For MATRIX-generated ReadAllGIC optimizations, the Gaussian route must use `Opt=ReadAllGIC`; for single-point jobs with a GIC geometry read, use the corresponding `Geom` form. MATRIX symmetrizes the GICs itself, so Gaussian-side automatic GIC symmetrization should not be requested.

4.2 Gaussian G16

```
setgau g16
g16run molecule.gjf
```

Use G16 when the job does not need GDV32-specific behavior. The same MATRIX adapter policy applies: logs, FCHK files and QFF data are copied back and then promoted through `matrix gaussian ...` commands.

4.3 Molpro

```
module load molpro/2025.3
molpro molecule.com
```

Molpro outputs enter MATRIX through the Molpro adapter. Downstream tools should consume the promoted `xyzin` sections and must not parse Molpro text privately.

4.4 ORCA

```
module load orca/6.1.1
orca molecule.inp > molecule.out
```

ORCA calculations can be launched on `oracle` for electronic-structure data. The MATRIX ORCA adapter currently promotes the final Cartesian geometry and, when ORCA prints a Cartesian Hessian, the shared `#CARTESIAN_HESSIAN` section used by GF/PED. Quantities not yet normalized into `xyzin` remain external data and must not be parsed privately by downstream scientific tools.

5 Background Submission

For production use, prefer the lightweight MATRIX submit helpers installed in `~/matrix/bin`. The source version of the submit helper is stored in the MATRIX repository as `scripts/remote/oracle/matrix-submit`. It creates one working directory per job, records the native program output, exposes a stable log link in `~/matrix/logs` and keeps a small metadata file with the command and process id.

```
matrix-submit gdv32 ~/matrix/inputs/molecule.gjf
matrix-submit g16 ~/matrix/inputs/molecule.gjf
matrix-submit molpro ~/matrix/inputs/molecule.com
matrix-submit orca ~/matrix/inputs/molecule.inp
```

Monitor jobs with:

```
matrix-status
matrix-tail 20260628-101010-gdv32-molecule
matrix-tail ~/matrix/logs/20260628-101010-gdv32-molecule.log
```

`matrix-submit` is a background wrapper based on ordinary Unix process management. It is not a queueing system. It is appropriate for this private machine, but it does not implement resource accounting, priorities or node scheduling. Each job writes `metadata.txt`; the fields `log`, `native_output` and `stdout` distinguish the stable log link, the scientific program output in the work directory and any launcher stdout/stderr stream.

6 Local MATRIX Remote Commands

The preferred local interface is the `MATRIX` remote QM wrapper. It uses SSH and SCP only as transport, calls the same remote `matrix-submit` helper and never stores credentials.

```
matrix qm remote-submit molecule.gjf --engine gdv32 --host enzo@oracle
matrix qm remote-submit molecule.com --engine molpro --host enzo@oracle
matrix qm remote-submit molecule.inp --engine orca --host enzo@oracle

matrix qm remote-status --host enzo@oracle
matrix qm remote-fetch JOB_NAME --host enzo@oracle --dest runs
```

`remote-fetch` reads the remote `metadata.txt`, copies the `native_output` file into a local job directory and writes `remote_qm_fetch_manifest.json`. When a safe adapter is already defined, fetching and promotion can be combined:

```
matrix qm remote-fetch JOB_NAME --host enzo@oracle \
  --dest runs --promote molpro --xyzin molecule.xyzin

matrix qm remote-fetch JOB_NAME --host enzo@oracle \
  --dest runs --promote gaussian-log-hessian --xyzin molecule.xyzin

matrix qm remote-fetch JOB_NAME --host enzo@oracle \
  --dest runs --promote orca --xyzin molecule.xyzin
```

`-promote auto` promotes Molpro outputs and ORCA outputs because those promotion paths are unambiguous. Gaussian promotion is deliberately explicit because a single log can contain different useful sections; choose `gaussian-log-hessian`, `gaussian-rovib`, `gaussian-electronic` or `gaussian-fchk`.

7 Copying Files

Manual copying remains a fallback. Copy inputs from the local workstation to the remote input directory:

```
scp molecule.gjf enzo@oracle:~/matrix/inputs/
scp molecule.com enzo@oracle:~/matrix/inputs/
scp molecule.inp enzo@oracle:~/matrix/inputs/
```

Submit remotely:

```
ssh enzo@oracle 'matrix-submit gdv32 ~/matrix/inputs/molecule.gjf'
ssh enzo@oracle 'matrix-submit molpro ~/matrix/inputs/molecule.com'
```

After the run, copy back the files needed by MATRIX adapters. The safest target is the `native_output` path recorded in `metadata.txt`; for Gaussian this is normally a `.log`, for Molpro a `.out` and for ORCA a `.out`.

```
scp enzo@oracle:~/matrix/jobs/JOB_NAME/molecule.log .
scp enzo@oracle:~/matrix/jobs/JOB_NAME/molecule.fchk .
scp enzo@oracle:~/matrix/jobs/JOB_NAME/molecule.out .
```

Use the actual job directory name printed by `matrix-submit` or listed by `matrix-status`. Keep large scratch files remote unless they are needed for a specific adapter or viewer.

8 Returning Results To MATRIX

The completed QM output must be normalized before GF/PED, MORPHEUS, VPT2/VCI, DVR, spectroscopy or GUI tools consume it. Typical local commands after copying files back are:

```
matrix gaussian promote-fchk molecule.fchk molecule.xyzin
matrix gaussian promote-log-hessian molecule.log molecule.xyzin
matrix gaussian promote-rovib molecule.log molecule.xyzin
matrix gaussian promote-electronic molecule.log molecule.xyzin
matrix molpro promote molecule.out molecule.xyzin
matrix orca promote molecule.out molecule.xyzin
```

For Gaussian ReadAllGIC optimization checks, keep the coordinate definition and the completed Gaussian result separate:

- the frozen MATRIX #GIC section is the authoritative coordinate source;
- the Gaussian input is only an external representation of those GICs;
- the Gaussian log supplies the final optimized geometry, Hessian and normal modes;
- GF/PED must rebuild B rows from the frozen MATRIX GIC definition on the final log geometry.

This avoids accidentally trusting a hand-edited or Gaussian-modified input block as the scientific coordinate definition.

9 Troubleshooting

Problem	Checks
Gaussian command not found	Run `setgau gdv32`, then `command -v gdv`. For standard Gaussian run `setgau g16`, then `command -v g16`.
Wrong Gaussian executable	Use only `gdv32` or `g16`. Do not use older GDV aliases.
Molpro unavailable	Run `module load molpro/2025.3` and `molpro -version`.
ORCA unavailable	Run `module load orca/6.1.1` and `command -v orca`.
No live output	Use `matrix-tail JOB_NAME` or inspect the log in `~/matrix/logs`.
Possible scratch/disk issue	Check `df -h ~ /local/scratch` and the program log.
MATRIX tool cannot use a result	Promote the QM output first and verify that the required `xyzin` section is present with `matrix validate` or the GUI readiness panel.

10 Operational Policy

- Keep `oracle` as an execution backend, not a parallel MATRIX development tree.
- Keep one adapter per external format. Gaussian, Molpro and ORCA outputs must be normalized before scientific tools consume them.
- Use `gdv32` or `g16`; do not revive older GDV variants.
- Use Molpro 2025.3 through `module load molpro/2025.3`.
- Use ORCA through the configured ORCA module.
- Keep credentials outside the repository and outside documentation.
- Keep job provenance: input file, MATRIX version, remote command, program version, log path and promoted `xyzin` sections.